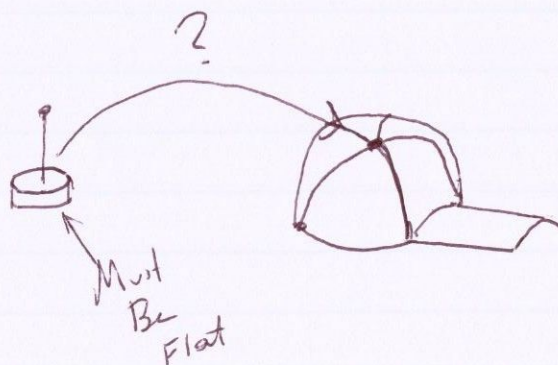
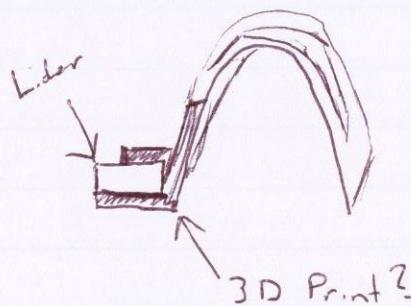
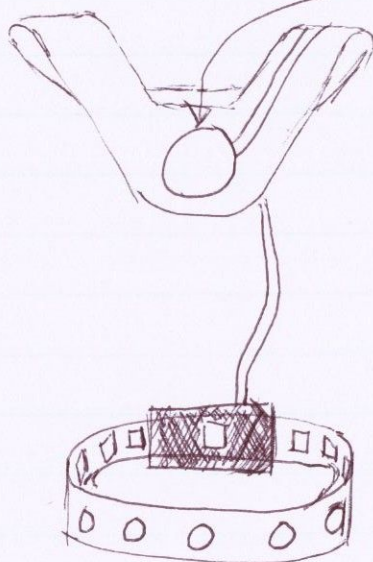
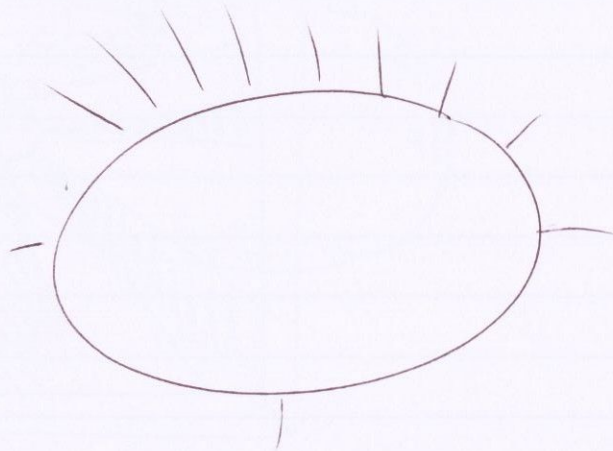
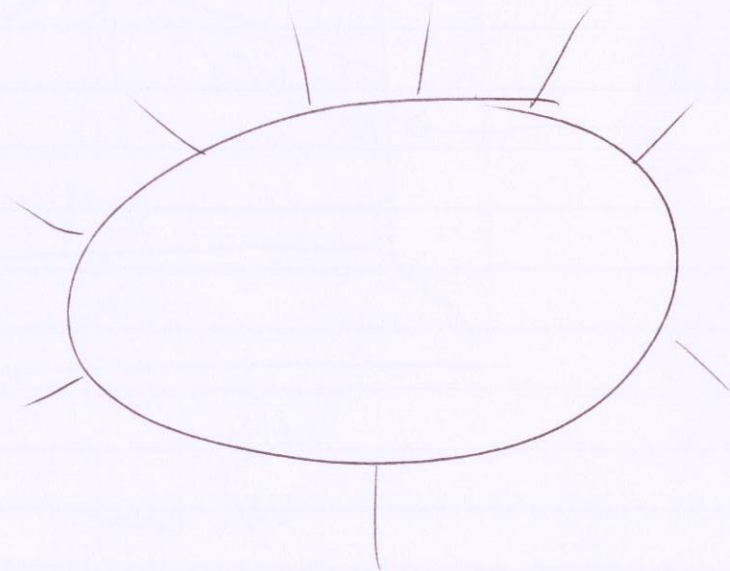
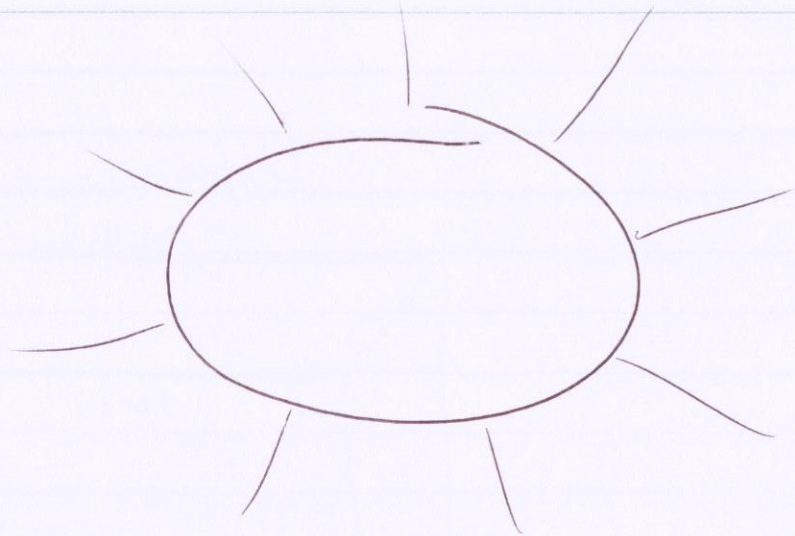


Doesn't  
Work

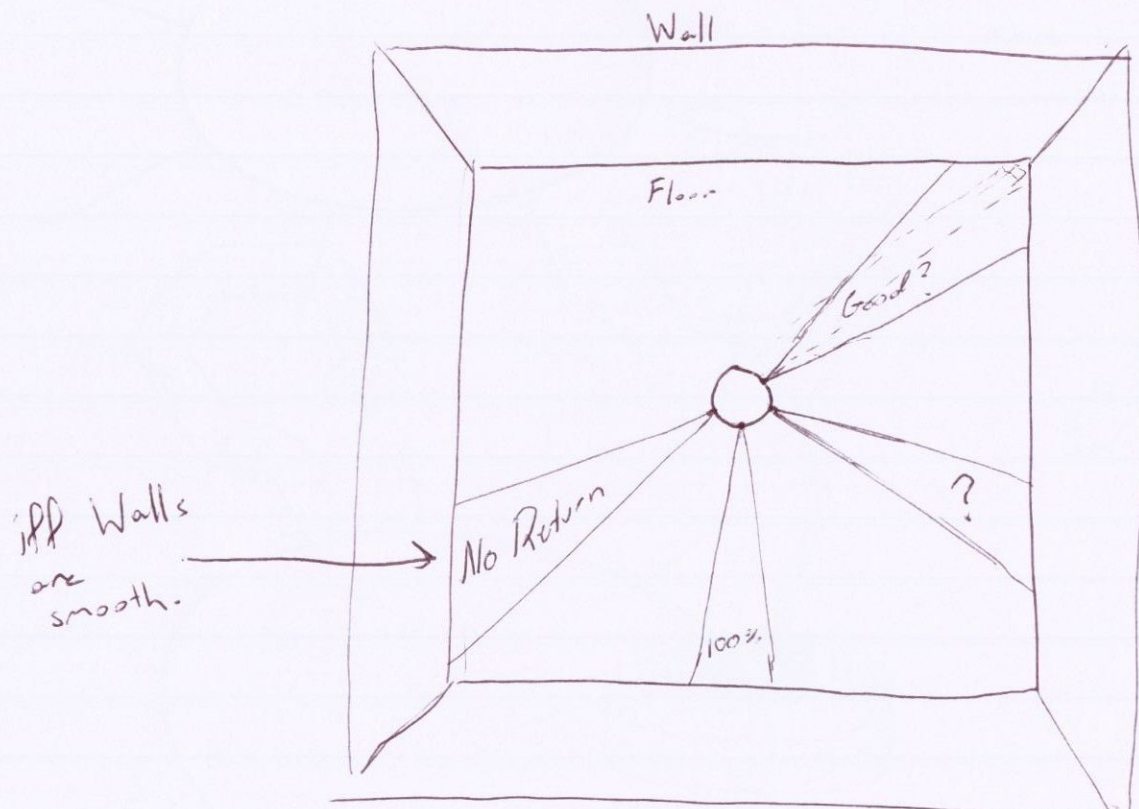


# HAPTIC Belt Design options

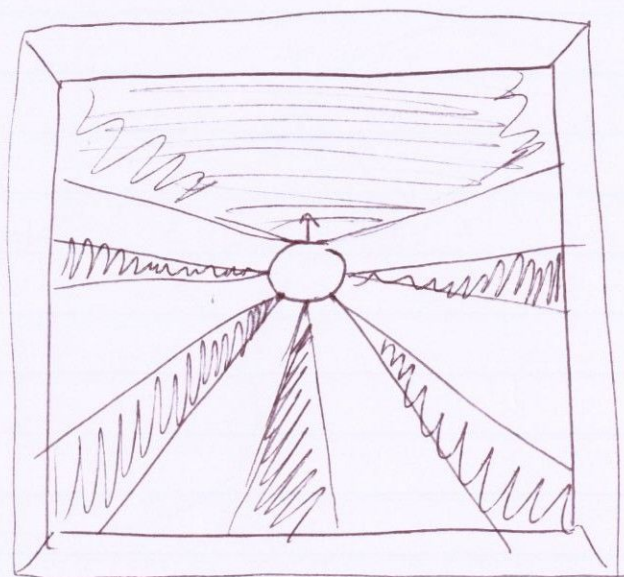


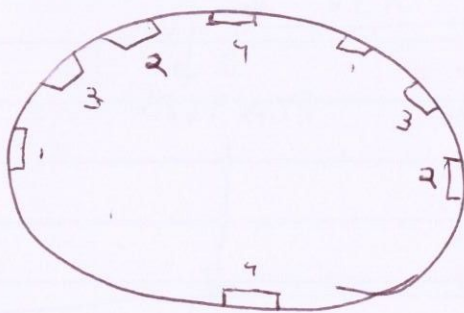


Ultrasonic only



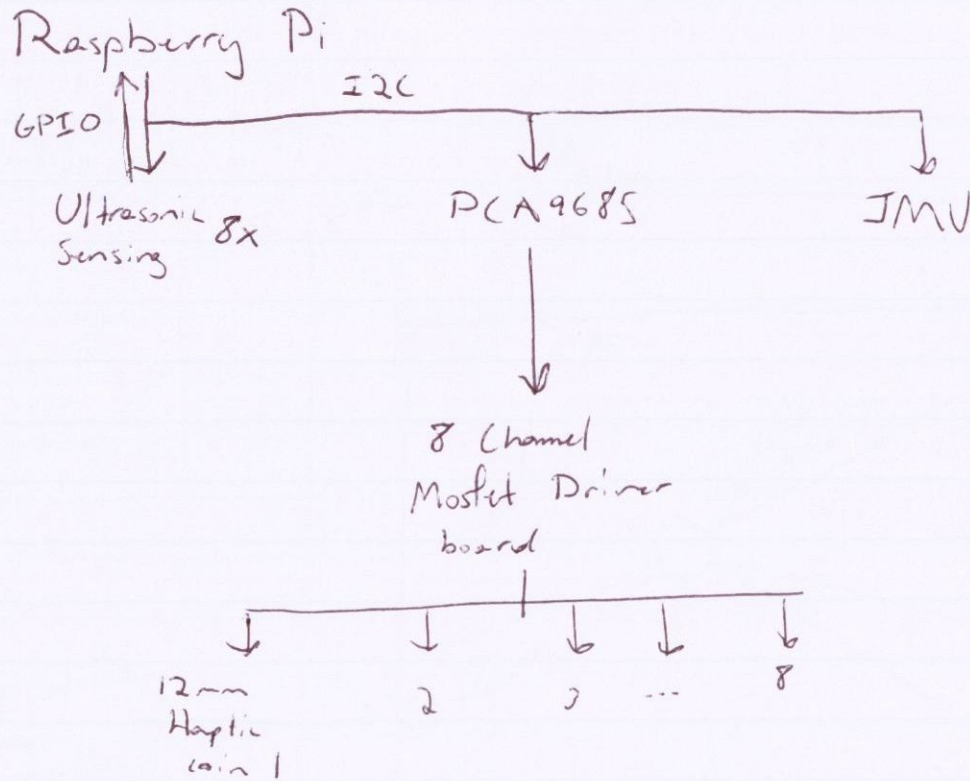
W/ Lidar



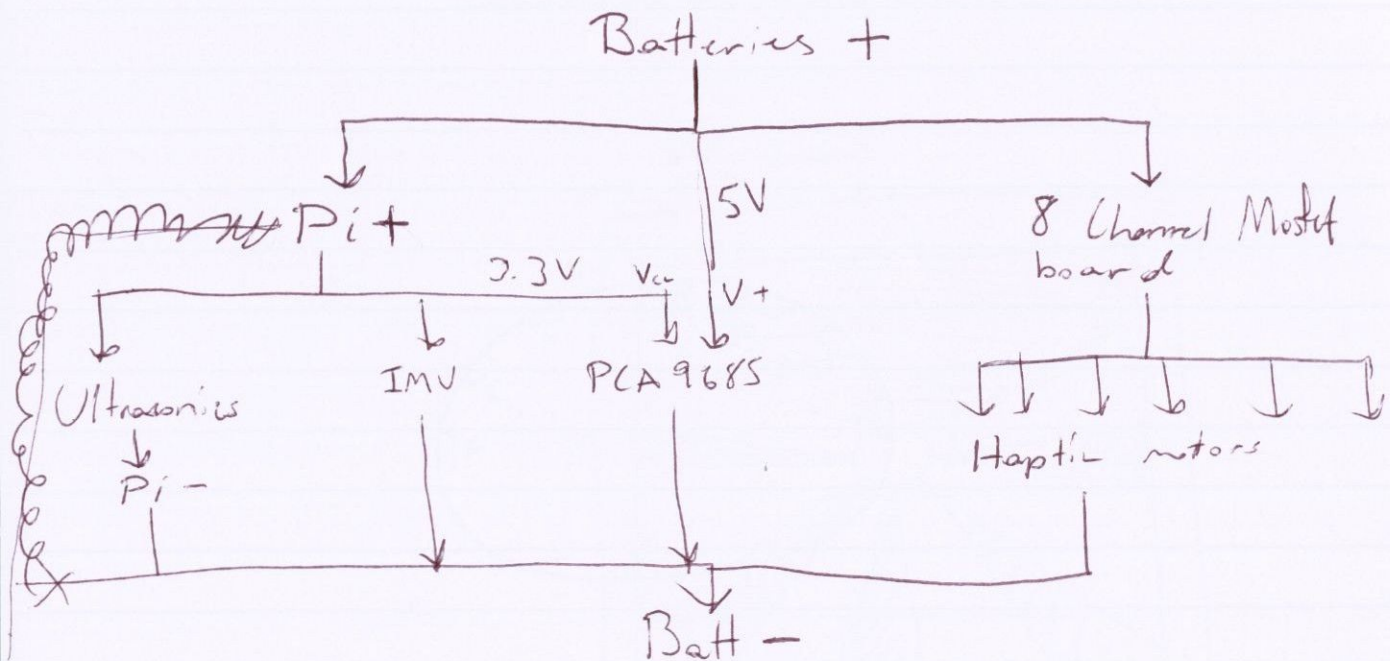




# One Line Diagram (Data)



# One Line Diagram (Power)



# Haptics W/ IMU

$$\text{IMU } \angle = x$$

$$\text{Hap } 1 \angle = x \pm 15^\circ$$

$$\text{Hap } 2 \angle = 30 + x \pm 15^\circ$$

$$\text{Hap } 3 \angle = 60 + x \pm 15^\circ$$

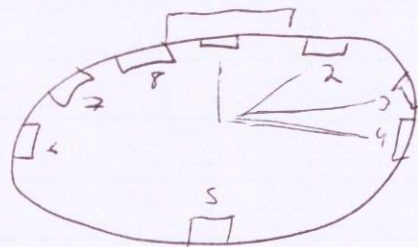
$$\text{Hap } 4 \angle = 90 + x \pm 15^\circ$$

$$\text{Hap } 5 \angle = 180 + x \pm 15^\circ$$

$$\text{Hap } 6 \angle = 270 + x \pm 15^\circ$$

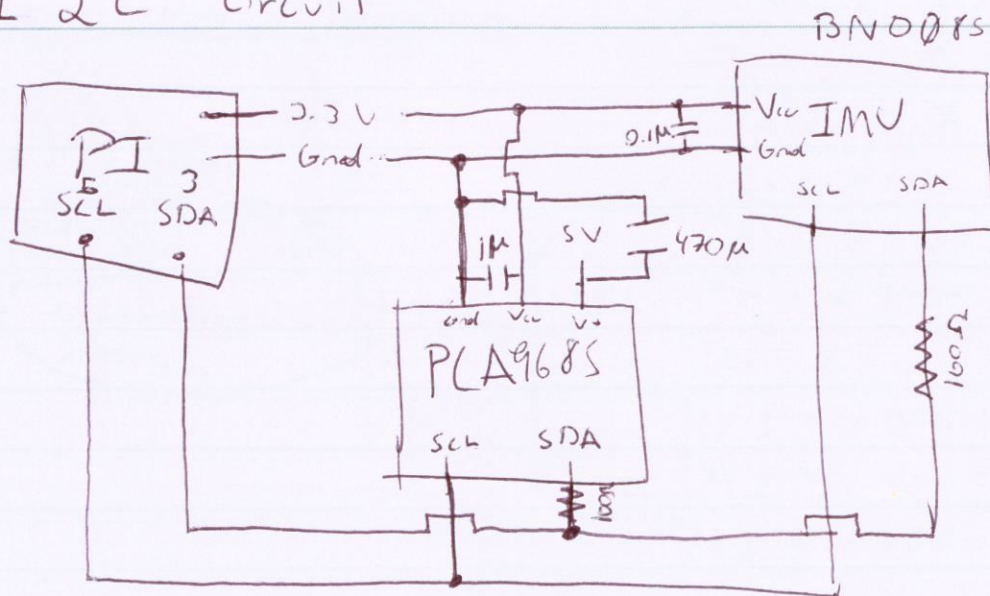
$$\text{Hap } 7 \angle = 300 + x \pm 15^\circ$$

$$\text{Hap } 8 \angle = 330 + x \pm 15^\circ$$

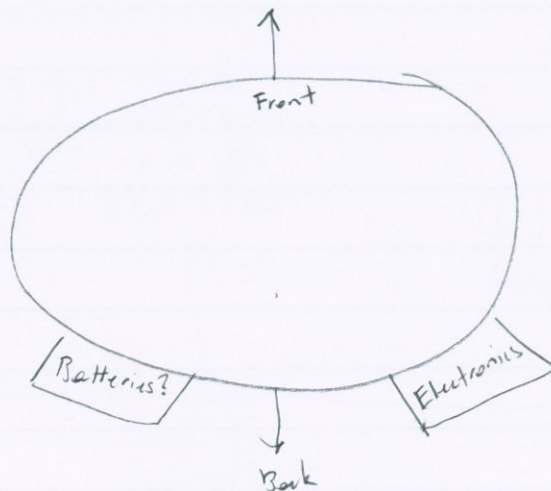
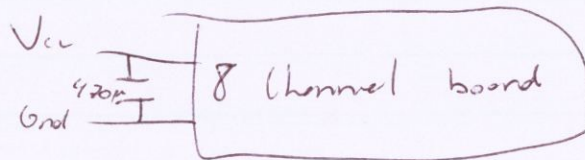




# I2C circuit



↑  
FAR  
↓  
AWAY

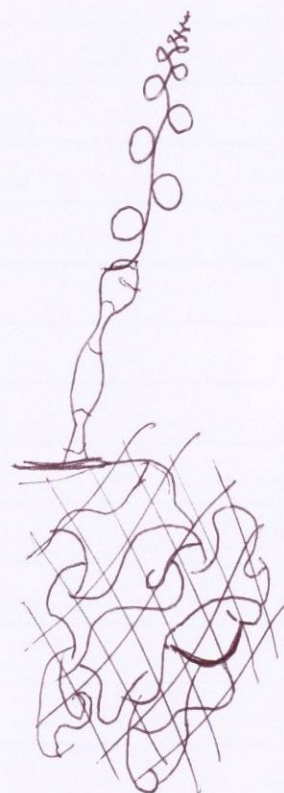


# PARTS LIST

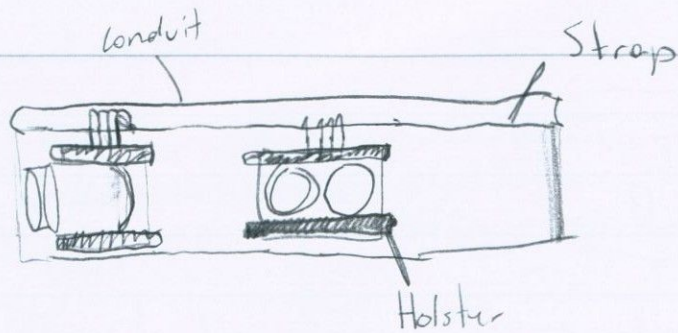
- PI
- 8x HC-SR04 Ultrasonic sensors
- BNO085 IMU
- 8x 12mm coin vibration motors
- PCA9685 Driver board
- 8-channel MOSFET Driver
- Power bank

0  
\$ 2.50 x 8 = \$20  
\$20  
\$ 2.00 x 8 = \$16  
\$8  
\$8  
0  
\$ 76

3  
70







For each sensor ←

Get distance + Direction + Position

if side sensor

if dist  $< 0.5$

Get last dist in direction / expected

if  $>$  error tolerance

Return

Else:

Get closest angle

Set distance for that angle

For each Vibrator

Get 2 closest angles

weighted average current data